Autonomous Robotic Ultrasound Vascular Imaging System With Decoupled Control Strategy for External-Vision-Free Environments

Guochen Ning, *Member, IEEE*, Hanying Liang, Xinran Zhang, and Hongen Liao, *Senior Member, IEEE*

Abstract*—*Objective: **Ultrasound (US) probes scan over the surface of the human body to acquire US images in clinical vascular US diagnosis. However, due to the deformation and specificity of different human surfaces, the relationship between the scan trajectory of the skin and the internal tissues is not fully correlated, which poses a challenge for autonomous robotic US imaging in a dynamic and external-vision-free environment. Here, we propose a decoupled control strategy for autonomous robotic vascular US imaging in an environment without external vision.** ***Methods:*** **The proposed system is divided into outer-loop posture control and inner-loop orientation control, which are separately determined by a deep learning (DL) agent and a reinforcement learning (RL) agent. First, we use a weakly supervised US vessel segmentation network to estimate the probe orientation. In the outer loop control, we use a force-guided reinforcement learning agent to maintain a specific angle between the US probe and the skin in the dynamic imaging processes. Finally, the orientation and the posture are integrated to complete the imaging process.** ***Results:*** **Evaluation experiments on several volunteers showed that our RUS could autonomously perform vascular imaging in arms with different stiffness, curvature, and size without additional system adjustments. Furthermore, our system achieved reproducible imaging and reconstruction of dynamic targets without relying on vision-based surface information.** ***Conclusion and Significance:*** **Our system and control strategy provides a novel framework for the application of US robots in complex and external-vision-free environments.**

***Index Terms*—Robotic ultrasound imaging system, vascular ultrasound imaging, ultrasound image segmentation, force control.**

I. Introduction

AUTONOMOUS robotic ultrasound imaging systems (RUS) offer advantages over freehand ultrasound imaging in terms of stability and reduced manual workload and have attracted much attention in recent years [1]. In some existing systems, RUS has been applied to various scenarios, such as autonomous vascular, spine, and liver imaging [2]. Although RUS has many specific structures and frameworks for different scanned targets, it still has many limitations and has not been widely used in clinical applications. In addition to the high cost, the complex clinical environment poses many challenges for reliable autonomous imaging processes. One of the major challenges is that the clinical imaging environment is unstructured, which limits the robust RUS control especially when external vision is disturbed. Furthermore, the spatial relationship between the skin and internal targets of the human body is uncertain in different subjects [3]. All these uncertainties mean the system and control processes still need to be adapted to different subjects even if the same target is imaged. Compared to freehand imaging, system adjustments are unacceptable in the clinical environment.

Vascular ultrasound imaging can provide valid diagnostic information on vascular distensibility, vascular stiffness assessment, and carpal tunnel syndrome, including vessel morphology, vessel wall thickness, and blood flow velocity [4], [5], [6]. Vascular ultrasound imaging usually relies on manual manipulation. Several studies have proposed the use of RUS for automated vascular scanning and 3D reconstruction [7], [8]. The RUS can be controlled in some simple imaging scenarios based on a standardized control framework and single-modality information. For example, the probe is easily contacted with the scanned surface, and there is little tissue deformation in the spine scanning task. Hase et al. and Li et al. focused on US image processing and guidance methods and proposed an image-based reinforcement learning-based (RL) RUS control framework for spine scanning [9], [10]. In their work, ultrasound image features guided the ultrasound probe to follow continuous spinal structures, but the skin-probe contact in the imaging process needed

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to be fully considered. Also focusing on scan direction and image guidance, Deng et al. proposed an ultrasound image-based control method for kidney scanning and verified it on a single phantom [11]. In their work, the imaging area was flat, and tissue deformation and surface-internal relationships were ignored. In some soft and more complex environments, Jiang et al. proposed a deformation-aware RUS for image deformation correction by contact force sensing [12]. Welleweerd et al. proposed an automated imaging method for breast imaging based on predefined imaging trajectories [13]. Although their imaging targets were more complex, the unstable contact caused by subject and target changes still needs to be thoroughly discussed.

To improve the unstable coupling of RUS with more complex surfaces, some researchers have integrated surface information into the control loop. Vision-based 3D surface reconstruction and scan path planning are typical approaches. Huang et al. reconstructed the entire scanned area with a depth camera and further outputted 6 degrees-of-freedom (DoF) control commands for full-path imaging [14]. Based on reconstructed 3D surfaces, Hennersperger et al. proposed to perform automatic ultrasound image acquisition in combination with MRI information [15]. While these works improved the interaction by reconstructing and imaging the entire area, scanning of areas outside the target was redundant. To reduce the time required for full-path planning, Kojcev et al. proposed a combination of 3D point clouds and manual trajectory to further filter scan paths further [16]. Ning et al. proposed an automatic RUS system based on RL and state representation learning to overcome the occlusion problem of visual methods [17]. Vision-based path planning can generate more complex commands for complex surface imaging, but the need for internal target information makes the scanning process to be redundant. In addition to visual information, the contact force is also introduced to control the probe posture. In [18], Tirindelli et al. integrated the relationship between ultrasound images and force information to improve the connection of the probe to the surface. In addition, Virga et al. used force information to correct the trajectory in RUS for 3D reconstruction [19]. Force information was fused with US images to perform targeted autonomous imaging commands [20]. Goel et al. proposed to collect the ultrasound image and the pose information at the probed point to estimate regions based on Bayesian optimization and hybrid force control [21]. The fusion of multimodality information has improved the accuracy and robustness. Surface modeling and path planning still correlate internal information with external information, which is also limited in the face of different subjects.

Some researchers have treated the ultrasound image information separately from the surface information to address more general application scenarios. In [22], Balter et al. proposed to use of stereo vision and ultrasound images for position and posture control. Spatial and posture information were decoupled to cope with different forearm morphologies. In further work, they integrated a near-infrared beam to improve the accuracy of vessel detection and the spatial positioning accuracy of the system [23]. Similar to the decoupling of the ultrasound probe control commands, Jiang et al. used the 3D point cloud to correct the spatial position of the probe instead of directly fusing the ultrasound image to the 3D surface for vascular imaging [24]. In addition, Victorova et al. used decoupled ultrasound image landmarks and contact force to guide robotic spinal imaging [25]. They used a PID method to maintain a specific 1-DoF angle between the probe and the back. In recent research, Wang et al. proposed to use of a 3D point cloud to generate a complete scan path to the breast, combined with a hybrid force-velocity control framework to perform a full coverage scanning process [26]. Based on 3D surface reconstruction and contact force information, Tan et al. also presented a dual-arms RUS for performing breast ultrasound scans [27]. Separating internal and external information seems straightforward, and dealing with the uncertain relationship between surface and internal tissue is more feasible. 3D reconstructed models were used to guide the posture of the probe and to decouple the posture control from the ultrasound image. The decoupled framework is a potential solution for complex imaging tasks and autonomous RUS. However, vision-based methods suffer from occlusion and the inability to detect tissue deformation because ultrasound scanning requires constant contact between the probe and the skin. 3D reconstruction-based methods are not feasible when external vision is unavailable or severely disturbed. The instability caused by the dynamic environment of automated imaging affects image quality and system safety, which must also be considered.

Rather than proposing a standard imaging framework, our goal is to improve the robustness of RUS for dynamic and uncertain targets. The solution focuses on decoupling the control strategy from the force-controlled trajectory generation. In manual ultrasound imaging, the operator scans by observing the ultrasound image and sensing the contact force after the probe contacts the body. The spatial position of the skin is usually not precisely located. Referring to the manual operation, we propose a decoupled control strategy to perform an autonomous robotic ultrasound vascular imaging process for different dynamic targets. The proposed strategy divides the imaging task into orientation and posture control, corresponding to the "*Where to go*" and "*How to go*" in manual operations. First, the location of the vessel in the ultrasound image guides the orientation of the ultrasound probe. To this end, we use a weakly-supervised segmentation network for vessel identification and localization, consisting of a transformer structure network with Doppler images as the label. For posture control, the contact force is used as guidance information and is decoupled from spatial information. Building on our previous work [37], we use an RL agent to establish the relationship between contact force and posture and to maintain a certain angle between the probe and the skin. Finally, the command from the two independent agents is integrated and executed through a force control method to ensure smooth interaction with the skin. Our main contributions are as follows: 1) With reference to manual operation, we propose a novel decoupling control strategy for environments without external vision. Probe orientation and posture are decided by two independent agents, including an image-guided deep learning (DL) agent and a force-guided RL agent. 2) Instead of presenting a 3D surface reconstruction framework, we propose to use a combination of ultrasound segmentation masks and contact

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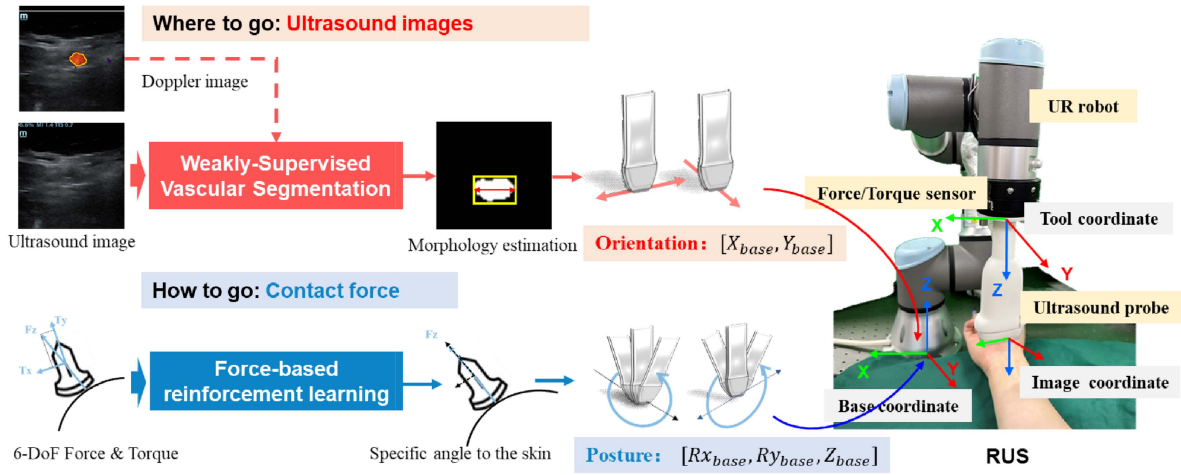


Fig. 1. General overview of the proposed control strategy. Automatic RUS imaging is divided into two independent control parts, including image-guided orientation control and force-guided posture control.

forces to improve the efficiency and robustness of the system. 3) The proposed control strategy and methods allow autonomous vascular imaging of different subjects while avoiding extra system adjustments. The imaging results of our system are feasible and stable and can also be used for reconstruction.

II. Methods

A. Control Strategy Overview

To cope with external-vision-free environments, we propose to decouple the skin surface from the internal tissues and further perform autonomous robotic vascular imaging. The framework of our control strategy is illustrated in Fig. 1. The imaging control process is divided into the outer-loop posture control and the inner-loop orientation control. In the orientation control, a weakly-supervised neural network model is used to segment and localize the vessel in the ultrasound images and to control the orientation of the probe. The 2-DoF command $T_{ori} \in \mathbb{R}^2$ in the orientation control includes the movement (X_{base},Y_{base}) in the base coordinate system. In the outer-loop control, the probe is maintained at a specific angle to the skin under the control of a force-guided RL agent for stable ultrasound image acquisition. The posture control command includes the roll angle (Rx_{base}), pitch angle (Ry_{base}) and Z-directional (Z_{base}) movement of the probe in the base coordinate system. Finally, the two parts of the commands are merged for complete imaging.

B. System Calibration Before Contact

Spatial calibration of the RUS is performed for image-based orientation guidance. There are three coordinate systems in our RUS, as shown in Fig. 1, including a Base coordinate system $Base$ for robot control, a tooltip coordinate system Tip for probe positioning, and an image coordinate system Img. There are the tip coordinate system calibration and the ultrasound image coordinate system calibration.

To acquire the posture of the ultrasound image, we conducted ultrasound calibration to determine the transformation matrix T_{Img}^{Tip} between the image coordinate system and the tooltip coordinate system. As shown in Fig. 2, a cross-wire phantom is mounted in the water tank for ultrasound calibration [28]. The cross-wire point was manually located by moving the robotic arm and detecting the point in the US images. To mitigate the effects of localization errors, the imaging procedure was repeated twenty times and imaging data were recorded from different positions and orientations. We set the origin of the phantom coordinate system Ph at the cross point for convenience, and the pixel (u, v) at the center of the cross-wire point should satisfy the transformation relation as:

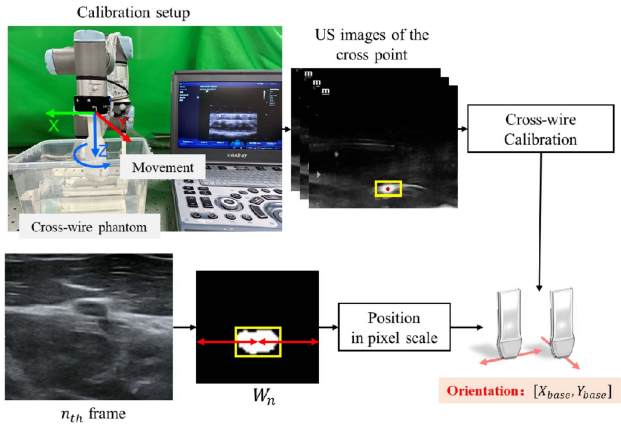


Fig. 2. Ultrasound calibration with a cross-wire phantom, based on which the vessel pixel positions in the image coordinate system are mapped into the tool coordinate system for orientation control of the US probe.

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = T_{Base}^{Ph} \cdot T_{Tip}^{Base} \cdot T_{Img}^{Tip} \cdot \begin{pmatrix} s_x u \\ s_y v \\ 0 \\ 1 \end{pmatrix} \quad (1)$$

where s_x, s_x are scale factors of the ultrasound images. The transformation matrix T_{Tip}^{Base} can be calculated directly according to the robot system configurations. The robust Levenberg-Marquardt algorithm iteratively optimizes the parameters [29].

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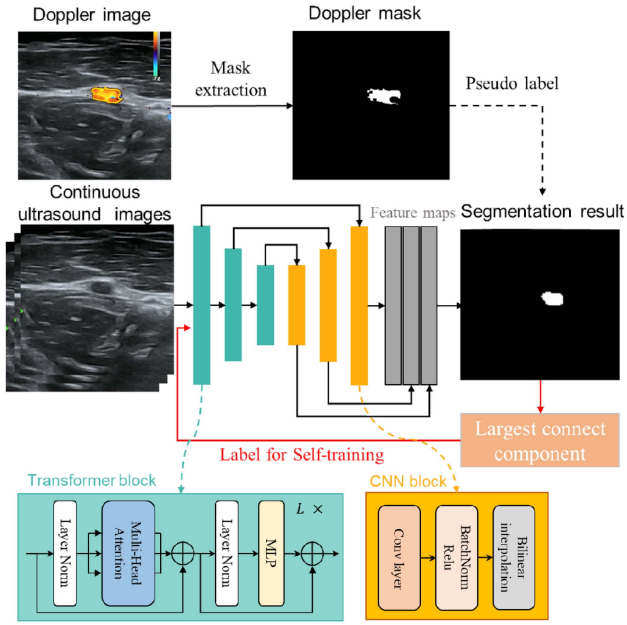


Fig. 3. Weakly-supervised vessel segmentation network based on Doppler images. The Doppler image is used as the pseudo label.

Tooltip coordinate system calibration involves determining the relationship between the tooltip coordinate system and the base coordinate system, which can be done accurately according to a basic calibration procedure [12]. Since the ultrasound images and tooltip positions are recorded under static conditions, additional time calibration is not required. In addition, the initial position of the arm in the workspace is determined manually.

C. Weakly-Supervised Vascular Segmentation and Orientation Guidance

The principle of Doppler imaging of blood vessels is to measure blood flow velocity using the Doppler effect. In contrast to B-model ultrasound images, Doppler images can provide morphologic and spatial information about the vessels and further guide vascular diagnosis. However, Doppler images can vary and be unstable as the heart beats. Compression of the vessel by the probe can also cause temporary closure of the vessel and further lead to failure of Doppler-based guidance. In addition, Doppler imaging requires the probe to be angled toward the vessel, which can be difficult in a dynamic environment. To obtain vessel guidance information, we decided to segment and identify vessels using Doppler images as segmentation labels to reduce the time consumption of vessel labeling time.

Transformer structure-based segmentation networks have demonstrated advanced global feature extraction capabilities and have been widely used in various image segmentation applications [30]. Based on [31], we use a Transformer encoder for the global image feature extraction and a convolutional neural network (CNN) decoder for upsampling, as shown in Fig. 3. The input $I_{us}^{W\times H\times n}$ to the network is n continuous ultrasound images stacked in the channel dimension to enhance the contextual information in the sequence, where (W, H) is the image size. After I_{US} is split into m image patches of size p, the feature sequence $f_m \in \mathbb{R}^{d\times m}$ with dimension d can be generated by a linear layer after flattening the image patches, where $m = (W/p) \times (H/p)$. The global dependencies between the image patch are captured by the Vision Transformer structure [32]. Feature vectors are positional-embedded by a learnable vector $E_{pos} = [pos_1, \ldots, pos_n] \in \mathbb{R}^{(HW/P^2)\times d}$ to retain the position information in this procedure, which can be formulated as:

$$Z_0 = f_m + E_{pos} \tag{2}$$

In the encoder, each Transformer block consists of L layers of multi-head self-attention blocks. Further, each self-attention model consists of three linear layers representing intermediate features with queries $\boldsymbol{Q} \in \mathbb{R}^{n\times d}$, keys $\boldsymbol{K} \in \mathbb{R}^{n\times d}$ and value $\boldsymbol{V} \in \mathbb{R}^{n\times d}$:

$$\text{MSA}\ (\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)\boldsymbol{V} \tag{3}$$

The network is employed by the multilayer perceptron (MLP) model. After the MLP model, a LayerNorm (LN) and residual connections are applied and connected [33]. Finally, Z_L is fed into an LN lay er which outputs the encoded features $Y \in \mathbb{R}^{P^2\times d}$:

$$\mathrm{Z}'_i = \text{MSA}\left(\text{LN}\left(Z_{i-1}\right)\right) + Z_{i-1},\ i = 1, 2\cdots L \tag{4}$$

$$Z_i = \text{MLP}\left(\text{LN}\left(Z'_{i-1}\right)\right) + Z'_{i-1},\ i = 1, 2\cdots L \tag{5}$$

$$Y = \text{LN}\left(\mathrm{Z}'_i\right) \tag{6}$$

In addition, a multi-level encoding structure containing three cascaded Transformer blocks is utilized for feature extraction from coarse to fine [31]. A 3 × 3 convolutional layer for feature map downsampling is contacted after each Transformer block.

To match to the multi-level structure of the encoder, a decoder consisting of three CNN blocks is used to upsample the feature vectors [34]. Each CNN consists of a 3×3 convolutional layer, a bilinear sampling operation, and a transform operation, and produces the multi-level feature maps $X_i \in \mathbb{R}^{d\times H\times W}$. After dimensionality reduction and Argmax operation, the segmentation map is output from the stacked feature map $X = [X_i | i = 1, \ldots m] \in \mathbb{R}^{(m\times d)\times H\times W}$ [34]:

$$X_i = \text{Bilinear}\left(H, W\right)\left(\text{Conv}\left(Y_i\right)\right), i = 1, 2\cdots, m \tag{7}$$

$$M_{class} = \text{Conv}\left(\text{Concat}\left(X_i\right)\right) \tag{8}$$

$$Mask = \text{Argmax}\left(M_{class}\right) \tag{9}$$

The loss function consists of a segmentation loss $\mathcal{L}_{seg}$ between the possibility map and the label, and a structure loss $\mathcal{L}_{NRDice}$ between the segmentation mask and the label:

$$\mathcal{L} = \mathcal{L}_{seg} + \mathcal{L}_{NRDice} \tag{10}$$

In our RUS, Doppler images corresponding to vessel images can be acquired simultaneously during data acquisition. The segmentation loss function is the commonly used cross-entropy loss. The structure loss function is the noise-robust dice score to

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deal with the voids in Doppler images [35]:

$$\mathcal{L}_{NRDice} = \frac{\sum_i^{HW} |\, p_i - g_i|^{\tau}}{\sum_i^{HW} p_i^2 + \sum_i^{HW} g_i^2} \tag{11}$$

where $p_i \in \mathbb{R}^{HW}$ and $g_i \in \mathbb{R}^{HW}$ denote the value of each pixel in the segmented mask and Doppler mask. The hyperparameter $\tau \in [1, 2]$ is the accuracy of Doppler masks for balancing the pseudo labels.

Labeling unseen data with labeled data can effectively improve the results, which is called self-training [36]. Some Doppler masks in the dataset are missing due to vascular pulsations, so the corresponding pseudo-label needs to be corrected. To reduce the noise interference in the Doppler image and ensure that the largest vessels are identified for directional guidance, the segmentation result is processed by the largest connective component module and used as the label for the self-training.

In addition to the pulsation of the vessel itself, the pressure and the associated spatial position between the imaging plane and the vessel can also change the morphology of the vessel in the image. Therefore, to simplify the orientation estimation process and improve efficiency, the position of the segmentation center of the mask in the ultrasound image is directly used to guide the probe movement. Meanwhile, the Doppler and segmentation masks are combined to guide the direction and improve the feasibility. According to the transformation matrix from the image coordinate system to the tool coordinate system in Section B, the pixel values can be converted to the distance values of the probe.

D. Force-Guided Posture Control

Probe posture control affects contact with the scanned surface and determines the effectiveness of image segmentation and orientation guidance. In the absence of external visual information, the contact force is used as the only information to guide probe contact with the skin. Factors that affect the contact force during scanning include friction, surface deformation and pressure. We have found that conventional linear force controllers (such as the PID method) have significant limitations in RUS posture adjustment [37]. The performance of conventional linear controllers for force control relies on hyperparameter tuning according to the environment, which is effective in 3D reconstruction-based RUS [26]. However, conventional controllers cannot deal with the non-linear relationship between posture and perceived force in environments without external visual information. The already adjusted hyperparameters are difficult to adapt to the dynamic environment. Therefore, a learnable force-based control method is proposed for RUS in a soft environment [37].

The principle of our force-guided approach is to build an RL agent that learns a direct relationship between the probe posture and the force sensed, avoiding complex force analysis and parameter tuning. An RL process in a Markov decision process can be defined by the sets of states $\mathcal{S}$, the continuous action space $\mathcal{A}$, the stochastic dynamics $p(s_{t+1}|s_t,\ a_t)$, the reward function $\mathcal{R}\ (s,\ a) = r \in \mathbb{R}$ and the discount factor $\gamma \in [0, 1)$ [38]. In the system-environment interaction at time step t, the expected reward is maximized by the parameterized policy $\pi_\theta(a|s)$. In our RUS, the state for posture control $s = (F_x, F_y, F_z, T_x, T_y, T_z)^T$ is the sensed 6-DoF force and torque.

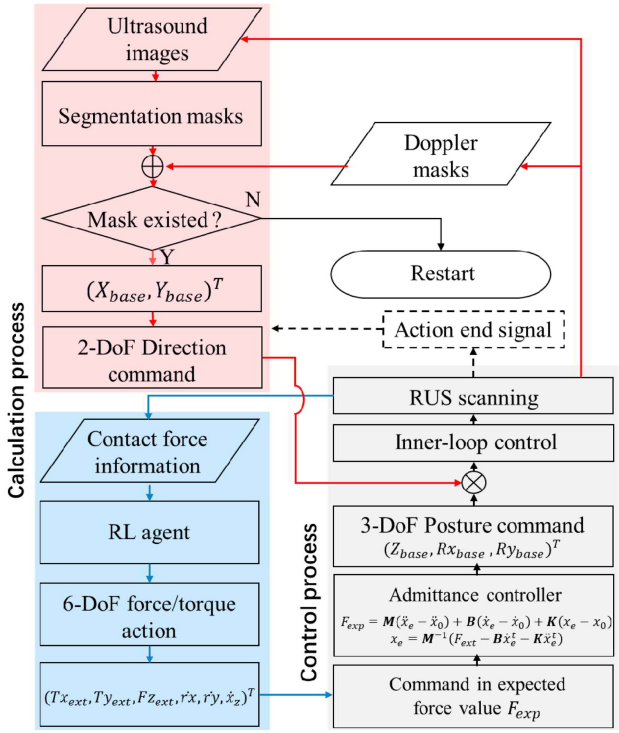


Fig. 4. RUS system logic. Force-based commands and image-based commands are generated in parallel and executed in sequence to ensure correct and smooth system logic.

As shown in Fig. 1, the 3-DoF posture control commands include roll angle (Rx_{base}), pitch angle (Ry_{base}) and Z-direction (Z_{base}) movement in a base coordinate system. This means that the image coordinate system is only rotated around the probe tip in the absence of image information. To execute the 3-DoF command, we propose a hybrid force-position control loop to combine and map the previous orientation and posture commands to the action space a_t, which will be introduced in the next section. The reward function R is a negative value of the square root of the torque in the $\mathcal{R}_x$ and $\mathcal{R}_y$ directions, which encourages minimizing the torque on the probe:

$$\mathcal{R} = \|T_x - (T_y - \tau)\| \tag{12}$$

where τ is a constant to adjust the angle of the probe in the sweep direction and to approximate the angle during the previous ultrasound image acquisition setup. The parameterised policy $\pi_\theta(a|s)$ is a multi-layer perceptron model with three hidden layers. Each hidden layer consists of a 256-dimensional fully connected layer and an activation layer. The training setup is posed in the framework of proximal policy optimization [39].

E. System Logic for Command Execution

As shown in Fig. 4, the decoupled control commands include an image-guided orientation command $(X_{base}, Y_{base})^T$ and a force-guided posture command $(Rx_{base}, Ry_{base}, Z_{base})^T$. The

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two agents are computed in parallel, and the generated commands are combined to ensure the smoothness of the imaging process. In the posture control framework, the characteristics of the skin (morphology, stiffness) are uncertain, which poses a challenge to the force control of the probe. We assume that the factors affecting the output force include the displacement and speed of the probe, which correspond to the force value and the control efficiency. Up to this point, it is natural to consider displacement and velocity as the action space of the posture. However, in our system, the spatial information on the skin surface is not available. Therefore, we use a force-displacement method based on admittance control to achieve a stable interaction with the skin [17].

In this method, the spatial variables of the probe (angle and displacement) are mapped onto the force space according to the admittance controller. Finally, the action space mentioned in the previous section is defined as the expected contact forces $(Tx_{ext}, Ty_{ext}, Fz_{ext})$ and corresponding velocities $(\dot{rx},\ \dot{ry}, \dot{x}_z)$ in the three relevant rotational directions [40]. According to the force-displacement method, the expected force and velocity output by the RL agent can be mapped to the rotation and displacement values:

$$F_{ext} = \boldsymbol{M}\left(\ddot{x}_{ext} - \ddot{x}_0\right) + \boldsymbol{B}\left(\dot{x}_{ext} - \dot{x}_0\right) + \boldsymbol{K}\left(x_{ext} - x_0\right) \tag{13}$$

$$x_{ext} = \boldsymbol{M}^{-1}\left(F_{ext} - \boldsymbol{B}\dot{x}^t_{ext} - \boldsymbol{K}\,\ddot{x}^t_{ext}\right) \tag{14}$$

These parameter matrices $[\boldsymbol{M}, \boldsymbol{K}, \boldsymbol{B}]$ are set to constant.

After combining with the orientation values, the complete control command is fed into the inner-loop orientation control and executed. The time consumption for each posture control step is uncertain. To ensure the correctness of the system control logic, the computation process at time step t is performed in parallel with the command execution at time step $t-1$. This is because image processing requires more computation time and resources. The following command is only executed after the previous action has been completed. With the control logic shown in Fig. 4, image errors do not lead to posture errors, because posture control is only guided by force. This means that posture control can be performed regardless of the presence or absence of ultrasound images. The posture of the probe determines the state of the ultrasound image. To dissociate orientation from posture, we used a segmentation network to identify and localise vessels in the image and estimate the direction. Posture errors, such as failed vessel identification, can still lead to image errors. In response, the system stops and restarts when a vessel identification error occurs.

III. Experiments and Results

A. System Setup and Implementation Details

CNN Model Training Setup: The clinical radial artery (RA) dataset for DL agent training included ultrasound and corresponding Doppler images. The acquired dataset consists of 20 subjects manually scanned by two sonographers. The sonographer controlled the probe to maintain a specific angle to the vessel to perform the scan and acquire stable and continuous Doppler images. After a simple thresholding process, the images in the pseudo-colour region were extracted as Doppler masks. In addition, Doppler images that were completely missing during data acquisition due to cardiac pulsation were automatically discarded. A total of 15000 RA images were acquired, of which 20% were used for validation and 10% were manually labelled for testing. The training setup involved training 300 epochs with batch size 16 using the Adam optimiser with an initial learning rate of 0.01, decreasing by half to 0.0001 after every 20 epochs. To increase computational speed, the region associated with blood vessels in the acquired ultrasound images was selected and resized to 100×100.

RL model training setup: The training process of the RL agent was performed on the real RUS [37]. The agent established the relationship between the contact force and the probe posture through repeated contact with a known phantom. A total of 20000 timesteps of training were performed.

System setup: The proposed RUS consists of a lightweight 6-DoF UR3 robot (UR3, Universal robot, Denmark), a 6-DoF force sensor (FT300 Force/Torque Sensor, Robotiq, Canada) for force measurement. B-mode US images of blood vessels in the upper limb were acquired using a 2D linear probe (L12-4s, Mindray DC 6E II, Mindray, China) with a centre frequency of 9.8 MHz and delay-and-sum (DAS) beamforming technique. The depth, spacing, and dynamic range of the US images were set to 1.5 cm, 0.046 mm and 110 dB, respectively. The working area and output force of the RUS were limited to ensure safety. The safety threshold for contact force is 15 N. All network calculations were performed on a workstation (Intel Core i7-8700K CPU, NVIDIA RTX 3080 GPU).

All data collection and experiments involved in this paper were approved by the local IRB (No. 20220152).

B. Segmentation and Orientation Guidance Evaluation

First, we verify the segmentation accuracy of the method (Our) under noisy labels. The test objects include test sets in the dataset and five continuously acquired ultrasound image sequences. Each sequence consists of 200 images. The metric used to evaluate the segmentation accuracy was DICE. In the ablation experiment, we first evaluated the effects of the number of encoder-decoder layers (Our-*n* layers), the number of continuous input images (Our-*n* images), the self-training procedure (Our-*w/o*-SelfTraining), and the NRDice loss (Our-*w/o*-NRDice) on the segmentation accuracy.

Compared to single image input (78.6±7.8%), the evaluation results of three continuous images (79.6±5.5%) indicated that the introduction of contextual information improved segmentation accuracy and was more applicable to continuous image sequences (83.3±4.1%). This conclusion was also valid for the evaluation results of the number of encoder-decoder layers (82.3±4.1%). The improvement of the segmentation results with more image inputs and network layers was small but led to an increase of the parameters. Therefore, we chose a compromise setting for our RUS (3 images, five layers). Both the self-training procedure and the NRDice loss were effective in improving the accuracy. Some qualitative segmentation results are shown in

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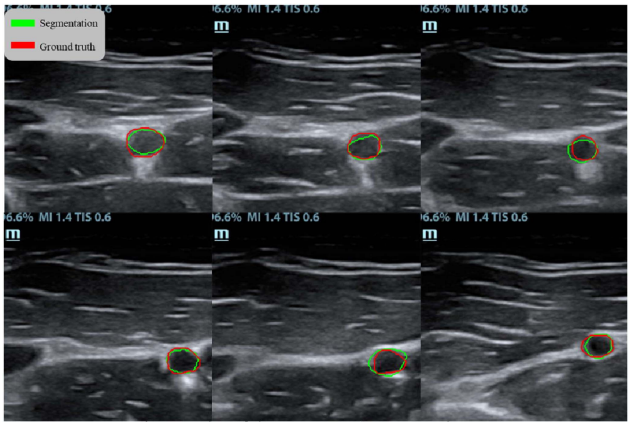


Fig. 5. Segmentation results of the segmentation network.

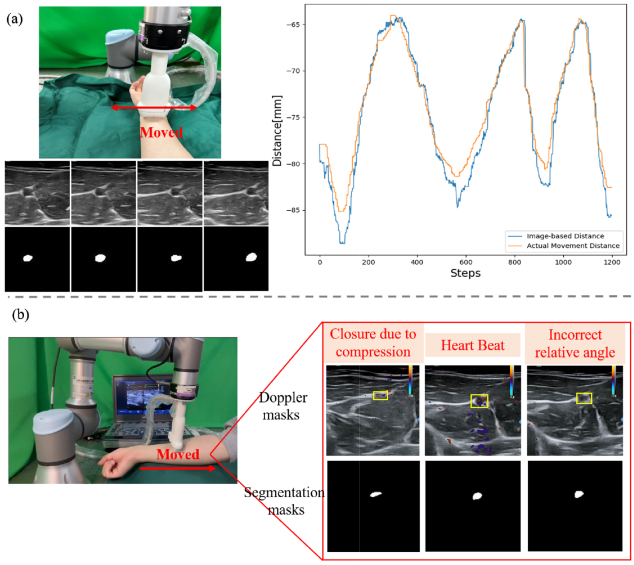


Fig. 6. Evaluation of the orientation guide. (a) RUS was moved according to the position of the segmentation masks in the image. The curve shows the difference between the actual displacement of the probe during movement and the displacement based on the segmentation results. (b) Vasoconstriction, compression, and improper angulation can lead to a failed Doppler acquisition.

Fig. 5. Since the segmentation mask was used here for guidance rather than for precise tissue delineation, this segmentation accuracy was considered feasible for our task.

In the orientation guidance evaluation, the RUS imaged a custom cubic silicone phantom with internal simulated blood vessels and was moved according to the position of the segmentation masks in the image. The mean error between probe displacement and vessel phantom displacement was 1.1±0.5 mm. The RUS was also moved in the y-direction and imaged the human arm as shown in Fig. 6(a). The mean error between the displacement obtained from the segmentation mask was 1.3±1.0 mm. The computation time for each ultrasound image segmentation was 0.02 s.

To further compare the performance of the segmentation mask and the Doppler mask as orientation guidance information, we performed five repetitive RUS imaging processes under the guidance of two types of information in the experiment. The segmented mask-guided RUS performed a complete scanning process in all five imaging processes, and quantitative details were demonstrated in the following experiments. In contrast, the Doppler mask-guided RUS failed to complete the imaging process due to compression and vasoconstriction, as shown in Fig. 6(b). Although Doppler masks were not available for various reasons, blood vessels in ultrasound images could still be segmented by neural networks and provided effective guidance information.

C. Image Acquisition and Posture Adjustment Evaluation

The purpose of posture control is to achieve correct contact with the skin without external visual information. To verify that convincing ultrasound images could also be obtained under force guidance, the RUS was only controlled by the outer loop part, i.e., the probe was centred on the tip (contact point) and rotated at the pitch and roll angles. There was no relative motion between the probe and the skin. For the static evaluation, a phantom was fixed in three different postures (Pot). The probe contacted four predefined positions (Pos) on the phantom in each posture and investigated the stable posture. The spatial position of the contact points is predefined in the robot coordinate system and could be repeated by RUS and manual operation. Each contact process was repeated five times to calculate the average value. The experiment used a phantom and a human arm as test subjects. The structural similarity (SSIM) between the acquired images of the same position was used as the evaluation metric.

Manually taken images at the same position were also used for comparison. The experimental procedure and quantitative results are shown in Fig. 7(a1) and (a2). The mean SSIM of all image pairs from repeated experiments and manual acquisitions in the same posture and position was 84.5±3.0% and 79.7±2.8%, respectively. The acquired images had high SSIM when the probe contacted the same points on the phantom in different postures. This indicated that the posture control effectively achieved a repeatable relative angular relationship with the contact surface under force guidance. The average SSIM of images taken at the same position between different postures was 79.8±2.8%. Comparison with manual control showed that posture control could achieve the correct angular relationship with the contact surface.

In a further dynamic evaluation, the RUS acquired images of the human arm. As with the static evaluation, only the posture of the ultrasound probe was adjusted with the arm. The probe did not scan after contact with the skin and there was no friction between the probe and the skin. To achieve this, the subject controlled their arm to maintain a fixed contact relationship with the probe. As shown in Fig. 7(b), the posture of the probe was adjusted with the movement of the arm after the contact point was reached. The arm then stopped moving and the probe followed to maintain a stable posture. The SSIM of the images acquired in the initial and stable postures was calculated to evaluate the effect of the arm movement on the reproducibility of the image acquisition. This process has performed a total of 10 times to obtain better statistics for performance evaluation. The average

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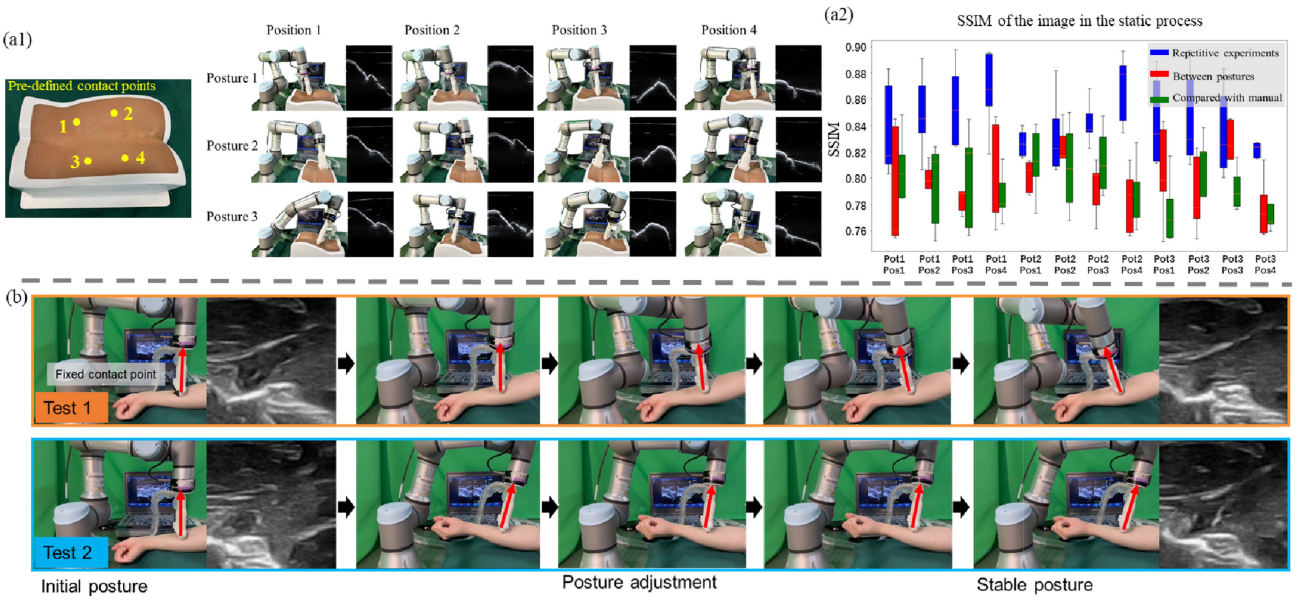


Fig. 7. (a) SSIM results of images acquired by the probe at the same position of phantoms with different postures under force control. (b) The posture of the probe was adjusted by moving the arm after reaching the contact point. The SSIM of the images acquired in the initial and the stable postures were calculated.

SSIM of the images acquired after the probe reached different stable postures was 49.7±8.5%. Although there was a decrease compared to the static repeat results, the same human structures could still be identified in the image. Overall, the SSIM of images acquired by RUS at the same position showed that our system could maintain a correct angular relationship with the arm during static and dynamic testing. This further ensured the reproducibility of the acquired images and the feasibility of using them for orientation guidance.

To evaluate the efficiency of the posture control process, a custom cubic silicone phantom with internal simulated blood vessels was placed in the workspace at a specific and known angle. As shown in Fig. 8, the probe quickly reached a perpendicular posture to the phantom after contact with the surface. Due to the surface deformation, the probe did not remain stationary after reaching the angle, but performed decreasing oscillations like a sine wave. Finally, the probe oscillated with one amplitude around the angle. After the change reaches stability (the variance tends to stabilise), the mean value of the angle of the probe is taken as the phase of the oscillation and is defined as the stable angle. In the repeated tests, the average time taken for a 10° gap was 0.9 s and the average time taken to reach stability was 4.0 s. As with the definition of time, the angular error was the error between the phase of the sinusoidal oscillation and the phantom angle. The error between the stabilized angle of the probe and the actual phantom angle was 0.45±2.72°. The significant peak in some of the shallower red lines was the amplitude of the sinusoidal oscillation. A large gap caused the RL agent to produce an overshoot value.

The results of each case represent each contact and control process, and are shown in Table II. The results show that the probe could quickly reach an angular gap of 10° and remained in a range of oscillations. As each contact caused different

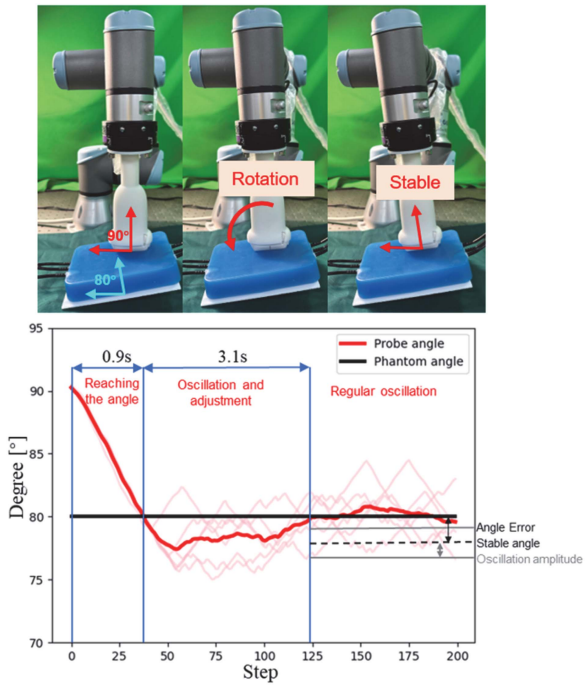


Fig. 8. Posture control efficiency for 10° gap.

compression and deformation of the phantom, the error and time results were also different. The larger error and longer adjustment time shown in Case 5 were due to the phantom being slightly displaced by the contact. The adjustment time of up to 3.7 s was the time for the probe to reach stability with an error adjustment of 10°. In fact, the time for the probe to reach the correct angle was 0.9 s, followed by an error oscillation adjustment of about 2–3°. Firstly, an error of 2–3° can meet the

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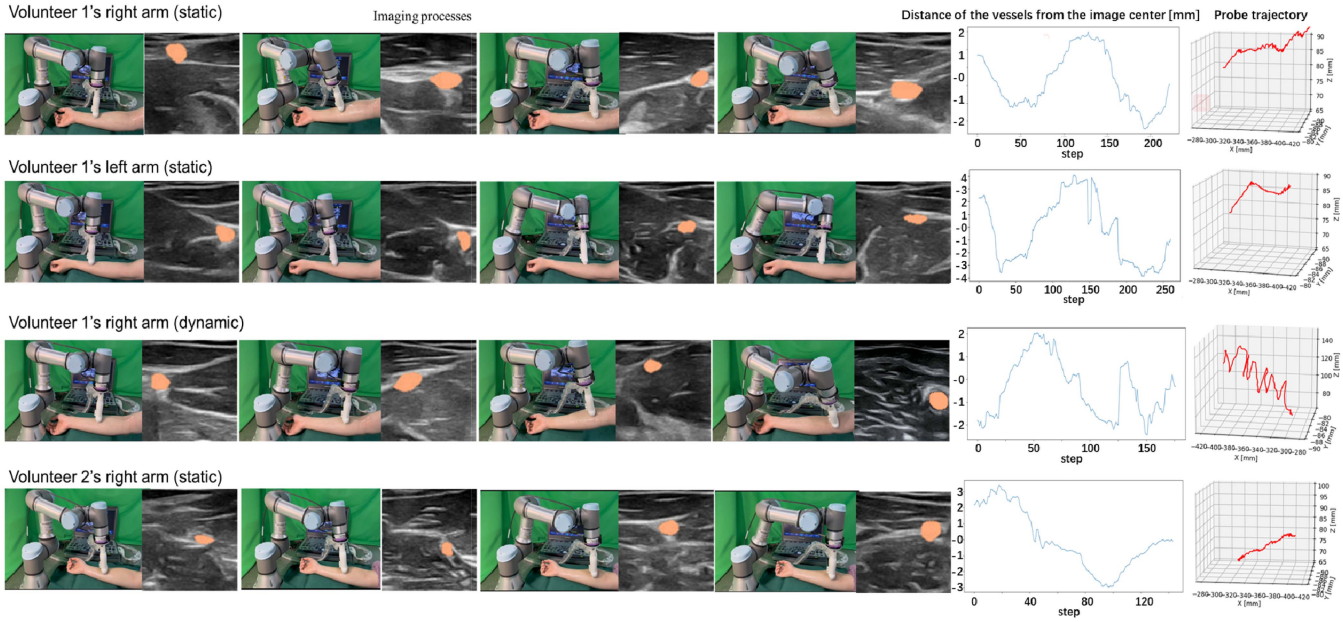


Fig. 9. Fully autonomous robotic vascular imaging procedures. Test subjects included static and dynamic left and right arms of different volunteers. The figure shows the position of the vessel in the ultrasound image and the trajectories of the probe during the imaging procedures.

TABLE I
SEGMENTATION ACCURACY EVALUATION

Method \ Data	Random test images	Continuous image sequence
Our-1 images-5 layers	78.6±7.8%	79.2±7.6%
Our-3 images-5 layers	79.6±5.5%	83.3±4.1%
Our-5 images-5 layers	79.7±5.4%	83.5±4.3%
Our-3 images -3 layers	77.7±7.3%	78.7±7.0%
Our-3 images -5 layers	79.6±5.5%	82.3±4.1%
Our-3 images -7 layers	79.4±5.3%	81.8±4.0%
Our-*w/o*-SelfTraining	77.8±6.9%	78.4±6.2%
Our-*w/o*-NRDice	78.5±6.1%	80.6±5.7%

TABLE II
ADJUSTMENT TIME RESULTS AND ANGULAR ERROR FOR EACH CONTACT

Case	Researching the angle/s	Mean error after reached/°	Maximum error/°	Reach stability/s	Stabilization error/°
1	0.9	0.39±1.11	2.98	1.1	0.45±1.13
2	0.9	0.64±1.49	3.96	2.7	-0.24±1.31
3	0.8	-1.00±1.00	3.41	1.3	-0.89±0.90
4	0.9	0.08±1.76	3.48	1.3	0.72±1.74
5	0.9	-2.46±1.32	4.91	4.4	1.67±1.25
6	0.9	-0.61±2.53	4.43	1.9	-1.96±1.04

requirements of vascular ultrasound imaging. It also means that the probe can receive the next motion command before it has fully reached stability. And the adjustment time of 0.9 s to cope with the 10° difference can also cope with radical movements, especially as the control speed has been limited to ensure safety.

D. Performance in Human Vascular Imaging

Human vascular imaging was performed on the RUS with the complete control framework. As shown in Fig. 9, the RUS contacted the volunteer's left and right upper arms according to the initial position in Section B and was guided by the force information and ultrasound image for the vascular imaging task. The subjects in the experiment included one adult Asian male and one adult Asian female with upper arm lengths of 26 cm and 21 cm, respectively. In the static imaging test, the arm was naturally placed in the working space of the RUS.

In general, subjects do not move significantly during routine clinical ultrasound examinations. However, subject motion can affect the imaging quality and safety of RUS. In previous experiments, we have verified the robustness of inner-loop image guidance to motion. To demonstrate the advantages of our system, volunteers performed a significant motion in dynamic experiments. On the one hand, the movement can cover the daily ultrasound examination environment and, on the other hand, provide a reference for some specific scenarios. In the dynamic test, the arm was deliberately moved during the imaging process. Manual imaging procedures of the same vessels were also performed and used as a gold standard. Each trial was repeated three times. Spatial position, posture, probe contact force, and acquired images were recorded during the experiment.

First, the acquired images were evaluated to verify the basic feasibility of our RUS. A total of 24 sequences were obtained, each containing an average of 245 images. To demonstrate the validity of the acquired images, the ratio of the number of images with vessels in the sequence was defined as the acquisition success rate. The average number of images without blood vessels in the static sequence was 3.1, and the average number in the dynamic test was 3.6. Some parts of the vessels were temporarily occluded due to probe compression and heart pulsation, causing the segmentation method to fail.

To further evaluate the quality of the acquired ultrasound images, the mean confidence of the ultrasound image sequences was used as an evaluation metric based on the confidence map,

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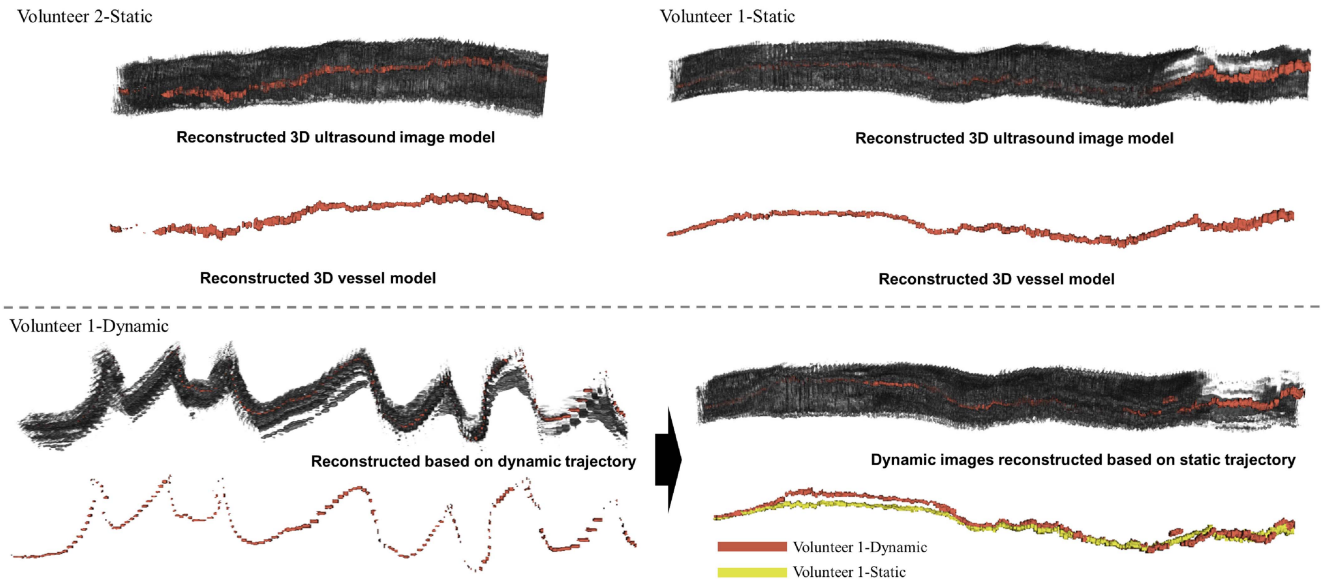


Fig. 10. Reconstructed 3D model based on acquired images and probe trajectories.

which indicates which part of the transducer area is in contact with the surface [41], [42]. As shown in Fig. 11, the upper part of the confidence map was selected to highlight the areas where the probe was in contact with the skin and to ignore shadows from the depths. The mean confidence of the ultrasound image sequences in the static test was 0.873. In comparison, the mean confidence of the dynamic test was 0.839. No probe detachment from the skin was detected in any of the acquired image sequences. The mean confidence of the manually acquired images was 0.912. Fig. 9 shows four autonomous imaging procedures, including dynamic, static, and two different subjects. In order to access the stability and accuracy of image guidance in the continuous acquisition, the mean distances of the vessels from the image center (vessel distance) were calculated in each sequence. In the static test, the mean vessel distance was 2.1±1.8 mm. Due to arm movement, the mean distance in the dynamic tests was 2.5±1.9 mm. The manually acquired images were more stable and had a mean length of 1.7±0.9 mm. The completeness of the acquired images showed that the level of coupling and the accuracy of posture adjustment were close to the manual level.

The contact force between the probe and the skin indicated the stability and safety of the system. In the static tests on two subjects, the average contact force of the probe was 10.3±2.3 N. In contrast, the contact force in the dynamic tests showed a significant variation (11.1±4.5 N). The maximum measured contact force was 15.3 N in a dynamic test when the subject moved the arm up in the imaging process. The quantitative evaluation results of the different ultrasound image sequences acquired are shown in Table II.

3D ultrasound volume reconstruction requires reproducible spatial surface information and ultrasound images in a dynamic environment. 3D point cloud reconstruction and registration are very effective for obtaining reproducible surface information. However, safe and reproducible dynamic imaging in environments without external vision is much more challenging. We used 3D Slicer [43] for model visualization and adjusted the voxel transparency of the volume to avoid occlusion and ensure better visualization of the vascular structures. As shown in Fig. 10, the acquired ultrasound image was reconstructed as a 3D model based on the previous calibration matrix and probe trajectory to demonstrate the repeatability and robustness of the proposed method. The hollow voxels appearing in the volume are caused by transparency adjustment for better visualization. In three 3D models from the static tests, vessels could be detected and reconstructed with a smooth continuous structure.

In comparison, during the dynamic test, the arm was moved up and down significantly, and the 3D model generated directly from the trajectory had to be corrected. According to the characteristics of the proposed decoupling control strategy, i.e., the spatial information was decoupled from the posture, and the image acquired in the dynamic test was reconstructed from the trajectory in the static test. The joint reconstruction results in the bottom right of Fig. 10 show a model-to-model distance of 0.832 mm compared to the vessel model reconstructed from the static scan. The result shows that our method can perform reproducible and safe RUS scanning even in unstable dynamic environments. *Several complete imaging sequences can be found in the supplementary video.*

E. Comparison With Vision-Based Coupled/Decoupled Control Strategy

The conventional RUS with coupled control strategy was compared with the proposed system built according to [14]. The 3D reconstruction system used for comparison had essential functions, including real-time arm reconstruction and positioning. A depth camera (RealsenseD457, Intel) was calibrated in the system and used for 3D model acquisition and reconstruction. We reconstructed the 3D surface and accurate predefined scan trajectories on the reconstructed model according to the

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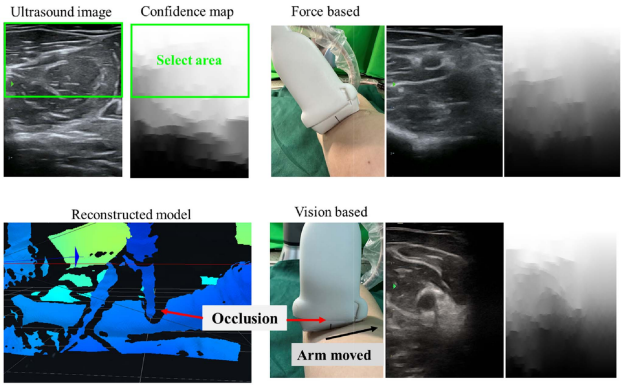


Fig. 11. Comparison with vision-based systems in dynamic tests. Due to occlusion caused by the probe, the probe did not make full contact with the skin due to non-reconstructable skin deformation.

morphology of the blood vessels. Based on the transformation matrix between the camera coordinate system and the robot coordinate system, the probe performed imaging under the control of 6-DoF commands.

In the static environment, the scan could be performed based on the reconstructed surface, although the deformation of the skin surface could lead to errors. The success rate, mean confidence, vessel distance, and contact force of the static arm imaging results for the static right arm of subject 1 were 100%, 0.875, 1.1±0.5 mm, and 10±0.5 N. The static comparison with the 3D reconstruction method showed that our approach could reach the level of this typical frame. However, the predefined trajectory compared to our method could not be directly transferred to the left arm of subject 1 or other subjects but had to be reconstructed and replanned.

Vision-based decoupling control strategies were also compared. Unlike the predefined trajectory, the dynamic probe trajectory was updated based on the real-time reconstructed 3D model. Compared to the predefined trajectories, the dynamic trajectories were able to cope with moving targets. The proposed system was designed for an environment without external vision, but the comparison would only be meaningful if full occlusion was present. Therefore, we chose a 3D reconstruction frame with minimal occlusion, where the probe occludes the contact points, for comparison. As shown in Fig. 11, when the arm moved in the direction of occlusion, the deformation of the skin could not be reconstructed due to the lack of a procedure to deal with deformation and occlusion, such as registration. The results in spatial positioning-based postures were unreliable when the arm was moved and resulted in incomplete contact between the probe and the skin. The evaluation metrics for subject 1's right arm were 88%, 0.710, 3.1±1.5 mm, and 14±3.7 N. In addition, the scanning paths had to be replanned for different subjects. In a dynamic comparison, these basic functions were not sufficient to cope with occlusion compared to our system.

IV. Conclusion

Automated robotic ultrasound systems face many challenges from clinical environments, such as uncertain imaging targets and dynamic environments. Herein, we propose an automated robotic vascular ultrasound imaging system with a decoupled control strategy for environments without external vision. In the proposed system, the "*Where to go*" and "*How to go*" of the probe correspond to the probe orientation and posture and are determined by two independent agents. First, the ultrasound image is used to guide the orientation of the probe. In the deep learning agent, we propose a weakly supervised segmentation method based on Doppler images to identify and localize blood vessels in ultrasound images. A transformer structure is introduced to improve the segmentation accuracy. We integrate and refine the system based on our previous work in the second reinforcement learning agent. Instead of conventional 3D reconstruction, contact force information is used to control posture and is decoupled from the ultrasound image.

Quantitative and qualitative experiments were performed to verify the feasibility of the proposed system. Image segmentation and orientation guidance were evaluated in the case of posture decoupling and achieved accuracies of 83.3 ± 4.1% and 1.3 ± 1.0 mm, respectively. In the evaluation of posture control, the probe achieved 79.8±2.8% SSIM for repeatedly acquired images of the same target without spatial information. Finally, our system performed complete vascular imaging processes on different arms of two volunteers for both static and dynamic situations. The experimental results showed that our system can perform vascular imaging on different subjects without additional adjustments or external visuals. The images acquired from the static and dynamic experiments could be reconstructed and recognized as 3D models. The results of the dynamic experiments demonstrate the robustness of our system to large movements, even when the subject's movements are unusual. The stable interaction and imaging performance of RUS in large and rapidly changing environments provides a reference for more complex environments, such as on-board RUS. Compared with basic conventional vision-based systems, our control strategy avoids the occlusion problem in vision systems and avoids the full-path scanning method to improve the imaging efficiency of the target.

In terms of limitations, our system cannot cope with the sudden movements of subjects. This is because the ultrasound images used for orientation guidance and contact force for posture guidance are assumed to be always available. When the arm is moved rapidly, a brief detachment of the probe from the skin causes both parts to fail. A potential solution would be to incorporate the stereo camera used for calibration into a real-time control loop to provide roughly rapid target localization.

Another limitation is that the vessels in the ultrasound images are not clear during the scan. In the case of multiple targets, correction of the identified blood vessels will be necessary. Fortunately, due to the continuity of vessel structure and simple morphology, imaging can still be done with other vessels tracked. Our future work will focus on more accurate target identification and segmentation. We will also attempt to image more challenging targets, such as abdomens with more complex tissues and discontinuous structures. In addition, system termination due to vessel identification errors must be considered in future work due to the lack of global information. We will focus

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on global information acquisition to improve the robustness of the system against errors. One potential solution is ultrasound image-preoperative image registration to obtain the relative spatial relationship of the current image to the arm.

Our control strategy shows that orientation control can be guided by more ultrasound image information by decoupling orientation from posture control. This means that our work has the potential to be extended to other imaging of different targets, such as abdominal organs, as long as valid ultrasound image guidance information can be obtained. Another potential area of future research is targeted posture control. In our work, we control the probe to remain at a specific angle to the skin to scan blood vessels. Clinical ultrasound diagnosis requires more complex control modes such as oscillation and rotation. Control modes other than sweeping are also worth developing. Overall, our system is feasible for autonomous robotic vascular ultrasound in uncertain environments and offers a potential solution for clinical applications.

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